DESCRIPTIVE ONLY - INTERNAL AUDIT SURFACE

Post-isolation multi-UAV descriptive evidence

Five independently trained seeds per arm, six prespecified scenarios, and 20 episodes per seed-scenario cell. This PDF is a portable view of the frozen descriptive summaries; the raw JSONL and validated JSON summaries remain authoritative.

Evidence boundary

No superiority, formal safety, component causality, calibrated uncertainty, real-time capability, or robust generalization claim is supported. The observed directions vary across scale, scenario, and metric, while collision outcomes often share a zero floor.

Aggregation contract

Each seed-scenario value first averages 20 episodes. Across-seed means and sample standard deviations use the five trained seeds as the repetition units. For 5/8 UAV, every paired delta is full uncertainty-aware minus independently trained no-uncertainty for the same seed.

Research mechanism under test

Communication staleness, delay, and loss -> neighbour-state prediction uncertainty -> uncertainty-aware predictive interaction graph -> coordinated action -> uncertainty-adaptive CBF margin -> dynamic-obstacle execution.

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3-UAV success by method and scenario

Across-seed descriptive means. All four independently trained arms and all six scenarios are shown. The self-loop graph is the historical raw-graph label, not a neighbour-state graph.

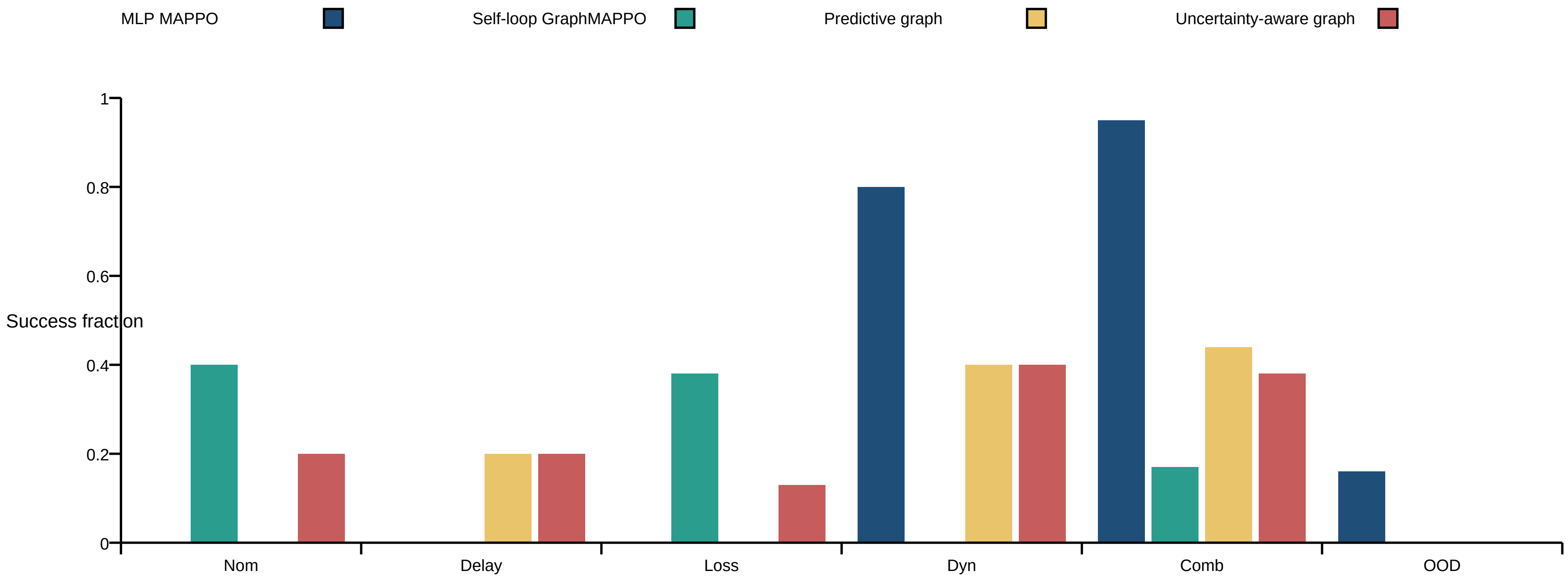


Chart labels: Nom = nominal; Delay = delay only; Loss = loss only; Dyn = dynamic only; Comb = combined; OOD = OOD communication plus obstacle.

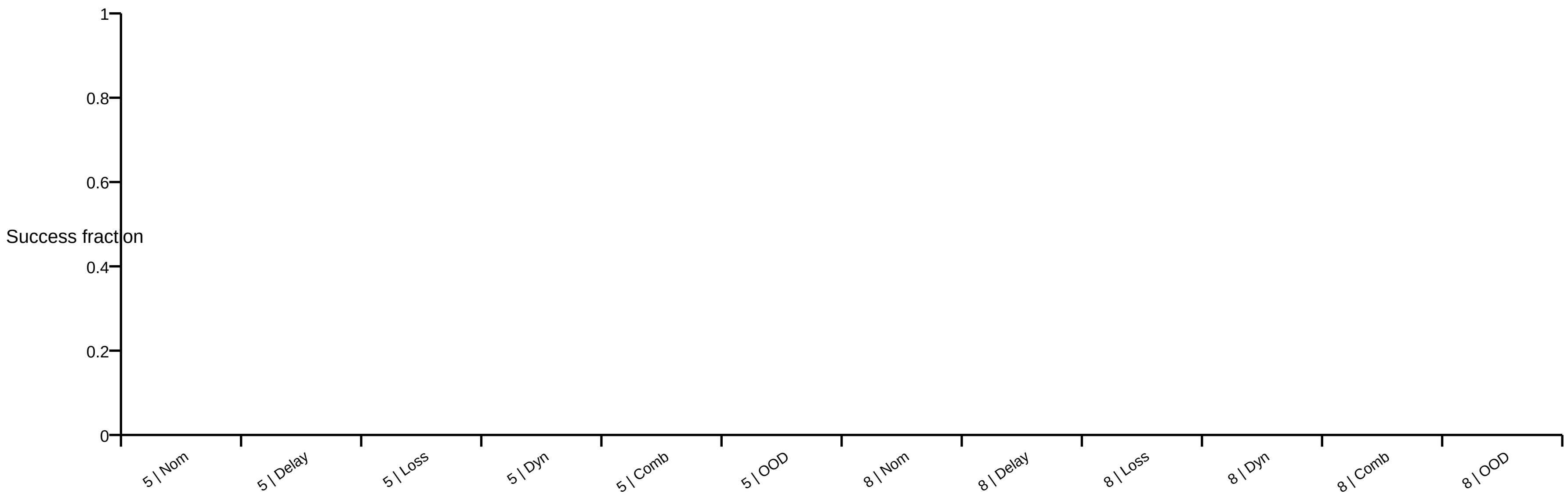
Exact 3-UAV success evidence

Mean and sample SD are across the five trained-seed cell means.

Scenario	Method	Mean	Seed SD
Nominal	MLP MAPPO	0.000	0.000
Nominal	Self-loop GraphMAPPO	0.400	0.548
Nominal	Predictive graph	0.000	0.000
Nominal	Uncertainty-aware graph	0.200	0.447
Delay only	MLP MAPPO	0.000	0.000
Delay only	Self-loop GraphMAPPO	0.000	0.000
Delay only	Predictive graph	0.200	0.447
Delay only	Uncertainty-aware graph	0.200	0.447
Loss only	MLP MAPPO	0.000	0.000
Loss only	Self-loop GraphMAPPO	0.380	0.522
Loss only	Predictive graph	0.000	0.000
Loss only	Uncertainty-aware graph	0.130	0.217
Dynamic only	MLP MAPPO	0.800	0.447
Dynamic only	Self-loop GraphMAPPO	0.000	0.000
Dynamic only	Predictive graph	0.400	0.548
Dynamic only	Uncertainty-aware graph	0.400	0.548
Combined	MLP MAPPO	0.950	0.112
Combined	Self-loop GraphMAPPO	0.170	0.380
Combined	Predictive graph	0.440	0.518
Combined	Uncertainty-aware graph	0.380	0.464
OOD comm. + obstacle	MLP MAPPO	0.160	0.358
OOD comm. + obstacle	Self-loop GraphMAPPO	0.000	0.000
OOD comm. + obstacle	Predictive graph	0.000	0.000
OOD comm. + obstacle	Uncertainty-aware graph	0.000	0.000

Full-method success by scale and scenario

The bars show only the full method's descriptive success context at 5 and 8 UAV. They are not a paired-effect plot. The exact paired table retains unfavorable cells, including 5 UAV delay_only (-0.20), 8 UAV delay_only (-0.40), 8 UAV loss_only (-0.09).



5/8-UAV paired success evidence

Both arm means, full-minus-no-uncertainty delta, and paired-delta sample SD are retained for every scale-scenario cell. The ablation jointly removes graph-risk and CBF-margin uncertainty and therefore cannot identify a component effect.

Scale / scenario	No-unc.	Full	Delta	Delta SD
5 UAV | Nominal	0.000	0.000	+0.000	0.000
5 UAV | Delay only	0.200	0.000	-0.200	0.447
5 UAV | Loss only	0.000	0.000	+0.000	0.000
5 UAV | Dynamic only	0.000	0.000	+0.000	0.000
5 UAV | Combined	0.000	0.000	+0.000	0.000
5 UAV | OOD comm. + obstacle	0.000	0.000	+0.000	0.000
8 UAV | Nominal	0.000	0.000	+0.000	0.000
8 UAV | Delay only	0.400	0.000	-0.400	0.548
8 UAV | Loss only	0.090	0.000	-0.090	0.201
8 UAV | Dynamic only	0.000	0.000	+0.000	0.000
8 UAV | Combined	0.000	0.000	+0.000	0.000
8 UAV | OOD comm. + obstacle	0.000	0.000	+0.000	0.000

Complete metric and provenance contract

The source summaries retain all 11 task metrics and five CBF diagnostic metrics for every eligible cell. Task outcomes and solver diagnostics remain separate; timing-sensitive replay results cannot replace the raw fallback ledger.

Task metrics	CBF diagnostic metrics
success collision terrain_violation threat_violation episode_return path_length mission_time minimum_separation temporal_conflict_count energy_proxy decision_latency	cbf_emergency_count cbf_emergency_fallback_rate cbf_intervention_rate cbf_mean_correction cbf_mean_solve_time_seconds

Limitations

Only five trained seeds are available and no inferential procedure was fixed before values were inspected. The uncertainty bound is deterministic rather than calibrated. The online CBF uses centralized simulator geometry, slack, discrete integration, and emergency fallback. Current evidence is therefore descriptive simulation evidence only.

Traceability

The adjacent provenance JSON records exact input paths and SHA-256 hashes, PDF hash, page count, Git revision, scenario set, visible row counts, and claim boundary. Eligibility and exclusions remain governed by docs/aamas2027_analysis_plan.md and docs/post_actor_isolation_descriptive_validation.md.